

Module-3

Discrete Random Variable

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Random Variables

- In analyzing random experiments, we typically focus on numerical aspects of the experiment.
- For example, in a soccer game, we might be interested in the number of goals, shots, or fouls. These quantities are examples of *random variables*.
- A random variable is a real-valued variable determined by an underlying random experiment.



Example

- Consider tossing a coin five times. The sample space S can be written as:

$$S = \{TTTTT, TTTTH, \dots, HHHHH\}$$

with $2^5 = 32$ elements. Let X represent the number of heads observed in this experiment. The random variable X can take values 0, 1, 2, 3, 4, 5, depending on the outcome.

- In essence, a random variable is a function that assigns a numerical value to each possible outcome of the experiment. For example, X assigns the value 0 to the outcome $TTTTT$, and the value 2 to the outcome $THTHT$.



Definition

- A random variable X is a function from the sample space S to the real numbers:

$$X : S \rightarrow \mathbb{R}$$

- We usually denote random variables by capital letters such as X , Y , and Z .
- The range of a random variable X , denoted by $\text{Range}(X)$ or R_X , is the set of possible values for X . In the above example, the range is:

$$\text{Range}(X) = R_X = \{0, 1, 2, 3, 4, 5\}$$



For the following random variables, we determine their ranges:

- **Problem 1**

I toss a coin 100 times. Let X represent the number of heads observed.

- **Problem 2**

I toss a coin until the first heads appears. Let Y represent the total number of coin tosses.

- **Problem 3**

The random variable T is defined as the time (in hours) from now until the next earthquake occurs in a certain city.



Solution

- The random variable X can take any integer from 0 to 100. Thus, the range of X is:

$$R_X = \{0, 1, 2, \dots, 100\}$$

- The random variable Y can take any positive integer, so the range of Y is:

$$R_Y = \{1, 2, 3, \dots\} = \mathbb{N}$$

- The random variable T can theoretically take any non-negative real number, so the range of T is:

$$R_T = [0, \infty)$$



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Discrete Random Variables

There are two important classes of random variables discussed:

- Discrete random variables (covered in this Slide)
- Continuous random variables (discussed in Next Slide)

A third class, called *mixed random variables*, which are a combination of discrete and continuous random variables, will also be discussed in next to next slide.



Countability of Sets

A set A is countable if:

- A is a finite set, e.g., $\{1, 2, 3, 4\}$
- A can be put in one-to-one correspondence with natural numbers (countably infinite)

Examples of countable sets:

- $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ and their subsets
- Non-empty intervals $[a, b] \subset \mathbb{R}$ are uncountable



Discrete Random Variables

A random variable X is called *discrete* if its range is countable.

Examples:

- In the previous example, the random variables X and Y are discrete.
- The random variable T is not discrete.

Thus, X is a discrete random variable if its range is countable.



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Probability Mass Function (PMF)

- For a discrete random variable X , the range R_X is a countable set.
- We can express the range as:

$$R_X = \{x_1, x_2, x_3, \dots\}$$

Here, $x_1, x_2, x_3, \dots$ represent the possible values of the random variable X .

- The probabilities of events $\{X = x_k\}$ are represented by the *Probability Mass Function (PMF)* of X .



Definition of PMF

- **Definition:** Let X be a discrete random variable with range $R_X = \{x_1, x_2, x_3, \dots\}$. The function:

$$P_X(x_k) = P(X = x_k) \quad \text{for } k = 1, 2, 3, \dots$$

is called the **Probability Mass Function (PMF)** of X .

- The PMF gives the probabilities of the possible values for a random variable.



Example-1

- **Example:** I toss a fair coin twice, and let X be the number of heads observed. Find the range of X , R_X , as well as its probability mass function P_X .



Solution

- The sample space is: $S = HH, HT, TH, TT$
- The number of heads X can take values 0, 1, or 2. Therefore, the range of X is: $R_X = 0, 1, 2$
- Next, we calculate the PMF of X :

$$P_X(k) = P(X = k) \quad \text{for } k = 0, 1, 2$$

- We have:

$$P_X(0) = P(X = 0) = P(TT) = \frac{1}{4}$$

$$P_X(1) = P(X = 1) = P(\{HT, TH\}) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

$$P_X(2) = P(X = 2) = P(HH) = \frac{1}{4}$$



Extending PMF to All Real Numbers

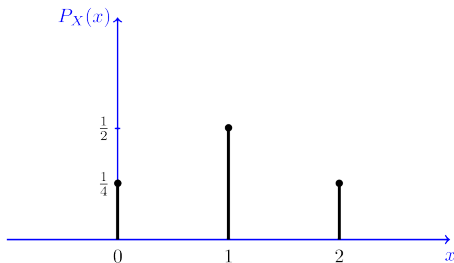
- Although the PMF is usually defined for values in the range R_X , we can extend the PMF to all real numbers:

$$P_X(x) = \begin{cases} P(X = x) & \text{if } x \in R_X \\ 0 & \text{otherwise} \end{cases}$$



Visualizing the PMF

- Figure shows the PMF for the random variable X . The random variable can take the values 0, 1, or 2, with $P(X = 1)$ being twice as likely as $P(X = 0)$ or $P(X = 2)$.
- If the experiment (tossing a coin twice) is repeated many times, about half the time we observe $X = 1$, and about a quarter of the time we observe $X = 0$ or $X = 2$.



Example-1

- I have an unfair coin for which $P(H) = p$, where $0 < p < 1$. I toss the coin repeatedly until I observe a heads for the first time. Let Y be the total number of coin tosses. Find the distribution of Y .



Solution

First, we note that the random variable Y can potentially take any positive integer, so we have

$$R_Y = \mathbb{N} = \{1, 2, 3, \dots\}.$$

To find the distribution of Y , we need to find $P_Y(k) = P(Y = k)$ for $k = 1, 2, 3, \dots$. We have

$$P_Y(1) = P(Y = 1) = P(H) = p,$$

$$P_Y(2) = P(Y = 2) = P(TH) = (1 - p)p,$$

$$P_Y(3) = P(Y = 3) = P(TTH) = (1 - p)^2 p,$$

$\vdots$

$$P_Y(k) = P(Y = k) = P(TT \dots TH) = (1 - p)^{k-1} p.$$

Thus, we can write the PMF of Y in the following way:

$$P_Y(y) = \begin{cases} (1 - p)^{y-1} p & \text{for } y = 1, 2, 3, \dots \\ 0 & \text{otherwise.} \end{cases}$$



Properties of PMF:

- 1 $0 \leq P_X(x) \leq 1$ for all $x \in R_X$.
- 2 $\sum_{x \in R_X} P_X(x) = 1$.
- 3 For any set $A \subseteq R_X$, $P(x \in A) = \sum_{x \in A} P_X(x)$.



Example-2

For the random variable Y in Example-1:

- Verify that $\sum_{y \in R_Y} P_Y(y) = 1$
- If $p = \frac{1}{2}$, find $P(2 \leq Y < 5)$



- Verify that $\sum_{y \in R_Y} P_Y(y) = 1$:

$$\sum_{y \in R_Y} P_Y(y) = p \sum_{k=0}^{\infty} (1-p)^k = p \frac{1}{1-(1-p)} = 1.$$

- If $p = \frac{1}{2}$, find $P(2 \leq Y < 5)$:

$$P(2 \leq Y < 5) = \sum_{k=2}^4 P_Y(k) = \frac{1}{2} \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{8} \right) = \frac{7}{16}.$$



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Independent Random Variables

- In real-life scenarios, we often deal with more than one random variable (e.g., weight and height of a person).
- We must consider not only each random variable individually but also their dependence (correlation).
- Two random variables are considered **independent** if knowing one does not affect the probabilities of the other.
- The definition for independent discrete random variables X and Y :

$$P(X = x, Y = y) = P(X = x)P(Y = y) \quad \text{for all } x, y.$$



Independent Random Variables (Cont'd)

- More generally, for independent random variables:

$$P(X \in A, Y \in B) = P(X \in A)P(Y \in B) \quad \text{for all sets } A, B.$$

- Intuitive understanding: X and Y are independent if knowing the value of one does not change the probability distribution of the other:

$$P(Y = y \mid X = x) = P(Y = y) \quad \text{for all } x, y.$$



Example: Coin Tosses

- Toss a coin $2N$ times.
- Let X be the number of heads in the first N tosses, and Y be the number of heads in the second N tosses.
- Since these are independent tosses, X and Y are independent random variables.



Example Problem

- Toss a coin twice, let X be the number of heads in the first two tosses, and let Y be the number of heads in the next two tosses.
- Find $P((X < 2) \text{ and } (Y > 1))$.



Example Problem

- Toss a coin twice, let X be the number of heads in the first two tosses, and let Y be the number of heads in the next two tosses.
- Find $P((X < 2) \text{ and } (Y > 1))$.
- Since X and Y are independent:

$$\begin{aligned} P((X < 2) \text{ and } (Y > 1)) &= P(X < 2)P(Y > 1) \\ &= (P(X = 0) + P(X = 1))P(Y = 2) = \left(\frac{1}{4} + \frac{1}{2}\right) \frac{1}{4} = \frac{3}{16}. \end{aligned}$$



Independence for n Random Variables

Consider n discrete random variables $X_1, X_2, X_3, \dots, X_n$. We say that $X_1, X_2, X_3, \dots, X_n$ are independent if

$$\begin{aligned} &P\left(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n\right) \\ &= P(X_1 = x_1)P(X_2 = x_2) \dots P(X_n = x_n), \quad \text{for all } x_1, x_2, \dots, x_n. \end{aligned}$$



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Special Distributions

- Some distributions are used frequently in practice and have been given special names.
- Each special distribution corresponds to a specific random experiment modeling real-life phenomena.
- Focus on understanding the random experiment behind each distribution, rather than memorizing Probability Mass Functions (PMFs).
- Key takeaway: Few new concepts, many formulas are variations of discrete random variable problems.



Bernoulli Distribution



Bernoulli Distribution

- The simplest discrete random variable can take only one value (non-random, not interesting).
- A Bernoulli random variable can take two possible values, typically 0 and 1.
- Models random experiments with two outcomes (e.g., "success" and "failure").



Examples of Bernoulli Distribution

- Pass-fail exam: Pass ($X=1$) or fail ($X=0$).
- Coin toss: Heads ($X=1$) or tails ($X=0$).
- Gender of a child: Male ($X=1$) or female ($X=0$).



Definition: Bernoulli Distribution

- A random variable X is said to be Bernoulli with parameter p , denoted as $X \sim \text{Bernoulli}(p)$.
- Its PMF is:

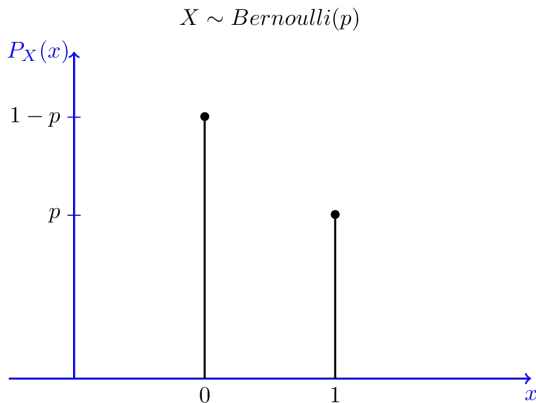
$$P_X(x) = \begin{cases} p & \text{for } x = 1 \\ 1 - p & \text{for } x = 0 \\ 0 & \text{otherwise} \end{cases}$$

- Where $0 < p < 1$.



PMF of Bernoulli Distribution

- The Probability Mass Function (PMF) of a Bernoulli random variable is shown graphically in Figure.



Bernoulli Random Variable and Event Association

- A Bernoulli random variable is associated with a certain event A .
- If event A occurs, $X = 1$ (e.g., passing a test), otherwise $X = 0$.
- This is why a Bernoulli random variable is also known as the **indicator random variable**.



Definition: Indicator Random Variable

- The indicator random variable I_A for an event A is defined as:

$$I_A = \begin{cases} 1 & \text{if event } A \text{ occurs} \\ 0 & \text{otherwise} \end{cases}$$

- I_A has a Bernoulli distribution with parameter $p = P(A)$, so:

$$I_A \sim \text{Bernoulli}(P(A)).$$



Geometric Distribution



Geometric Distribution: Random Experiment

- Consider a coin with probability $P(H) = p$.
- Toss the coin repeatedly until observing the first heads.
- Define X as the total number of coin tosses in this experiment.
- X follows a geometric distribution with parameter p .
- This experiment can be seen as repeating independent Bernoulli trials until the first success.



PMF of Geometric Distribution

- The range of X is $R_X = \{1, 2, 3, \dots\}$.
- The probability mass function (PMF) for X is:

$$P_X(k) = P(X = k) = (1 - p)^{k-1}p, \quad \text{for } k = 1, 2, 3, \dots$$

- Define $q = 1 - p$, so the PMF becomes:

$$P_X(k) = pq^{k-1}, \quad \text{for } k = 1, 2, 3, \dots$$



Definition: Geometric Distribution

- A random variable X is said to be a geometric random variable with parameter p , denoted as $X \sim \text{Geometric}(p)$.
- Its PMF is:

$$P_X(k) = \begin{cases} p(1-p)^{k-1} & \text{for } k = 1, 2, 3, \dots \\ 0 & \text{otherwise} \end{cases}$$

- Where $0 < p < 1$.

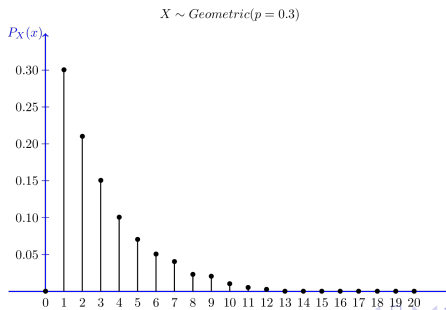


Geometric Distribution: PMF Variation

- Figure shows the PMF of a Geometric(0.3) random variable.
- Some books define the geometric random variable X as the total number of failures before the first success.
- In this case, the range of X is $R_X = \{0, 1, 2, \dots\}$, and the PMF is:

$$P_X(k) = p(1 - p)^k, \quad \text{for } k = 0, 1, 2, \dots$$

- In this book, $X \sim \text{Geometric}(p)$ means X is the total number of trials (Definition 3.5).



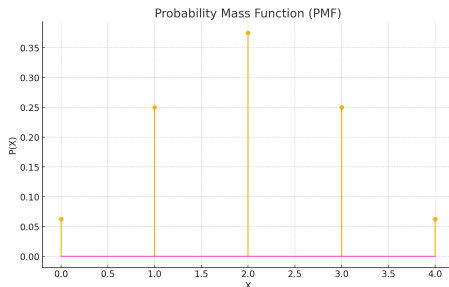
Binomial Distribution



Binomial Distribution: Random Experiment

Example

- Consider a coin with $P(H) = p$.
- Toss the coin n times, and define X as the total number of heads observed.
- Then $X \sim \text{Binomial}(n, p)$, with range $R_X = \{0, 1, 2, \dots, n\}$.
- If n is 4 then $X = \{0, 1, 2, 3, 4\}$ $P(x) = 1/16, 4/16, 6/16, 4/16, 1/16$



Definition: Binomial Distribution

- A random variable X is said to be binomial with parameters n and p , denoted as $X \sim \text{Binomial}(n, p)$.
- Its PMF is:

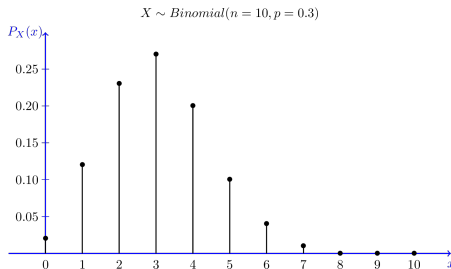
$$P_X(k) = \begin{cases} \binom{n}{k} p^k (1-p)^{n-k} & \text{for } k = 0, 1, 2, \dots, n \\ 0 & \text{otherwise} \end{cases}$$

- Where $0 < p < 1$.

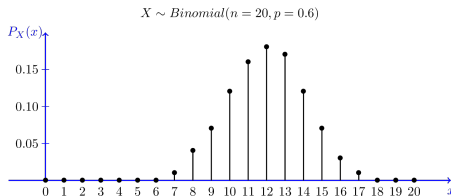


Binomial Distribution: PMF Examples

Binomial(10, 0.3) random variable



Binomial(20, 0.6) random variable



Binomial as a Sum of Bernoulli Variables

- A Binomial(n, p) random variable can be seen as a sum of n independent Bernoulli(p) random variables.
- **Lemma:** If $X_1, X_2, \dots, X_n$ are independent Bernoulli(p) random variables, then:

$$X = X_1 + X_2 + \dots + X_n \sim \text{Binomial}(n, p)$$

- To generate a random variable $X \sim \text{Binomial}(n, p)$:
 - Toss a coin n times.
 - Count the number of heads.
- This is equivalent to summing independent Bernoulli trials:

$$X_1 + X_2 + \dots + X_n$$

where each $X_i = 1$ if the outcome is heads, and $X_i = 0$ otherwise.



Example: Sum of Two Binomial Variables

Example 3.7

Let $X \sim \text{Binomial}(n, p)$ and $Y \sim \text{Binomial}(m, p)$ be two independent random variables. Define a new random variable as $Z = X + Y$. Find the PMF of Z .

Solution

Since $X \sim \text{Binomial}(n, p)$, we can think of X as the number of heads in n independent coin tosses, i.e., we can write

$$X = X_1 + X_2 + \dots + X_n,$$

where the X_i 's are independent *Bernoulli*(p) random variables. Similarly, since $Y \sim \text{Binomial}(m, p)$, we can think of Y as the number of heads in m independent coin tosses, i.e., we can write

$$Y = Y_1 + Y_2 + \dots + Y_m,$$

where the Y_j 's are independent *Bernoulli*(p) random variables. Thus, the random variable $Z = X + Y$ will be the total number of heads in $n + m$ independent coin tosses:

$$Z = X + Y = X_1 + X_2 + \dots + X_n + Y_1 + Y_2 + \dots + Y_m,$$

where the X_i 's and Y_j 's are independent *Bernoulli*(p) random variables. Thus, by Lemma 3.1, Z is a binomial random variable with parameters $m + n$ and p , i.e., $\text{Binomial}(m + n, p)$. Therefore, the PMF of Z is

$$P_Z(k) = \begin{cases} \binom{m+n}{k} p^k (1-p)^{m+n-k} & \text{for } k = 0, 1, 2, 3, \dots, m+n \\ 0 & \text{otherwise} \end{cases}$$



Negative Binomial (Pascal) Distribution



Negative Binomial (Pascal) Distribution

- The Pascal distribution generalizes the geometric distribution.
- Random experiment: Repeated independent trials until observing m successes.
- Suppose a coin has $P(H) = p$. We toss the coin until m heads are observed.
- Let X be the total number of coin tosses. Then $X \sim \text{Pascal}(m, p)$.
- The Pascal distribution can be written as $\text{Pascal}(1, p) = \text{Geometric}(p)$.
- Range: $R_X = \{m, m + 1, m + 2, m + 3, \dots\}$.



Deriving the PMF

- Event $A = \{X = k\}$ can be written as $A = B \cap C$, where:
 - B is the event that $m - 1$ heads are observed in the first $k - 1$ trials.
 - C is the event that a head is observed in the k -th trial.
- Since B and C are independent, we have:

$$P(A) = P(B)P(C)$$

- Using the binomial formula for $P(B)$ and $P(C) = p$, we obtain:

$$P_X(k) = \binom{k-1}{m-1} p^m (1-p)^{k-m}$$



Definition of Pascal Distribution

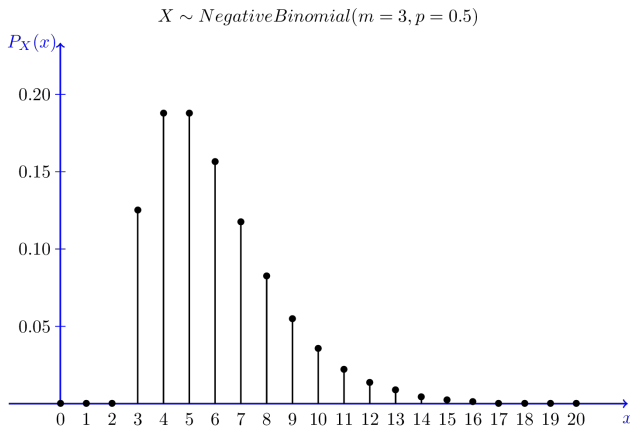
- A random variable $X \sim \text{Pascal}(m, p)$ has the PMF:

$$P_X(k) = \begin{cases} \binom{k-1}{m-1} p^m (1-p)^{k-m} & \text{for } k = m, m+1, m+2, \dots \\ 0 & \text{otherwise} \end{cases}$$



Example

Figure shows the PMF of a Pascal(m, p) random variable with $m=3$ and $p=0.5$



Hypergeometric Distribution



Hypergeometric Distribution

- Random experiment: Choose k marbles from a bag containing b blue and r red marbles without replacement.
- Let X be the number of blue marbles chosen. The PMF is:

$$P_X(x) = \frac{\binom{b}{x} \binom{r}{k-x}}{\binom{b+r}{k}} \quad \text{for } x \in R_X$$

- Range: $R_X = \{\max(0, k - r), \dots, \min(k, b)\}$
- Describes the probability of obtaining x blue marbles in the sample.



Poisson Distribution



Poisson Distribution

- The Poisson distribution is used for counting occurrences of events in an interval of time or space.
- Example: Counting the number of customers visiting a store from 1pm to 2pm.
- Suppose on average $\lambda = 15$ customers visit the store.
- The random variable X representing the number of customers can be modeled as $X \sim \text{Poisson}(\lambda = 15)$.



Definition of Poisson Distribution

- A random variable $X \sim \text{Poisson}(\lambda)$ has the range $R_X = \{0, 1, 2, 3, \dots\}$.
- The PMF is given by:

$$P_X(k) = \begin{cases} \frac{e^{-\lambda} \lambda^k}{k!} & \text{for } k \in R_X \\ 0 & \text{otherwise} \end{cases}$$



Validating the Poisson PMF

- We need to check that $P_X(k) \geq 0$ for all k and that $\sum_{k \in R_X} P_X(k) = 1$.
- Using the Taylor series for e^x , $e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$, we can verify:

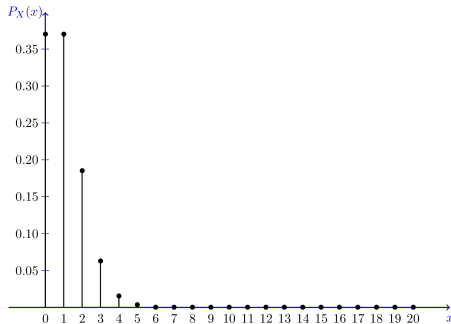
$$\sum_{k \in R_X} P_X(k) = \sum_{k=0}^{\infty} \frac{e^{-\lambda} \lambda^k}{k!} = e^{-\lambda} e^{\lambda} = 1$$

- Hence, the PMF is valid.

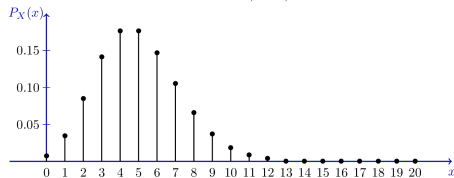


Poisson PMF Examples

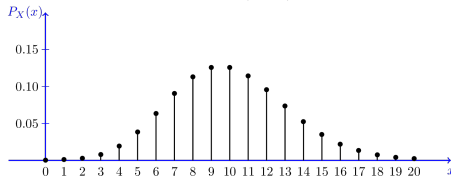
$$X \sim \text{Poisson}(\lambda = 1)$$



$$X \sim \text{Poisson}(\lambda = 5)$$



$$X \sim \text{Poisson}(\lambda = 10)$$



Example: Number of Emails in a Weekday

- The number of emails I receive on a weekday is modeled by a Poisson distribution.
- The average number of emails is 0.2 emails per minute.
- What is the probability that I get no emails in a 5-minute interval?
- What is the probability that I get more than 3 emails in a 10-minute interval?



Solution (Part 1)

- Let X be the number of emails received in the 5-minute interval.
- $X \sim \text{Poisson}(\lambda = 5 \times 0.2 = 1)$.
- The probability of receiving no emails in the 5-minute interval is:

$$P(X = 0) = \frac{e^{-\lambda} \lambda^0}{0!} = e^{-1} \cdot 1 = \frac{1}{e} \approx 0.3679$$



Solution (Part 2)

- Let Y be the number of emails received in the 10-minute interval.
- $Y \sim \text{Poisson}(\lambda = 10 \times 0.2 = 2)$.
- The probability of receiving more than 3 emails in this interval is:

$$P(Y > 3) = 1 - P(Y \leq 3)$$

- Using the Poisson PMF:

$$P(Y > 3) = 1 - (P_Y(0) + P_Y(1) + P_Y(2) + P_Y(3))$$



Solution (Part 2 Continued)

$$P_Y(k) = \frac{e^{-\lambda} \lambda^k}{k!}$$

$$P(Y > 3) = 1 - \left(e^{-2} + 2e^{-2} + \frac{4e^{-2}}{2!} + \frac{8e^{-2}}{3!} \right)$$

$$P(Y > 3) = 1 - e^{-2} \left(1 + 2 + 2 + \frac{8}{6} \right) = 1 - \frac{19}{3} e^{-2}$$

$$P(Y > 3) \approx 0.1429$$



Poisson Approximation for Binomial Distribution

- The Poisson distribution can be viewed as the limit of the binomial distribution.
- Suppose $X \sim \text{Binomial}(n, p)$, where n is large and p is small.
- Assume that $\lambda = np$ is a positive constant.
- We show that the PMF of X can be approximated by the PMF of a $\text{Poisson}(\lambda)$ random variable.
- This is useful since the Poisson PMF is easier to compute than the binomial PMF.



Theorem

Let $X \sim \text{Binomial}(n, p = \frac{\lambda}{n})$, where $\lambda > 0$ is fixed. Then, for any $k \in \{0, 1, 2, \dots\}$, we have:

$$\lim_{n \rightarrow \infty} P_X(k) = \frac{e^{-\lambda} \lambda^k}{k!}$$



Proof (Step 1)

We start with the binomial PMF:

$$P_X(k) = \binom{n}{k} p^k (1-p)^{n-k}$$

Substitute $p = \frac{\lambda}{n}$:

$$P_X(k) = \binom{n}{k} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k}$$

Take the limit as $n \rightarrow \infty$:

$$\begin{aligned} \lim_{n \rightarrow \infty} P_X(k) &= \lambda^k \lim_{n \rightarrow \infty} \left[\frac{n(n-1)(n-2) \dots (n-k+1)}{n^k} \right] \\ &\quad \cdot \lim_{n \rightarrow \infty} \left(1 - \frac{\lambda}{n}\right)^n \cdot \lim_{n \rightarrow \infty} \left(1 - \frac{\lambda}{n}\right)^{-k} \end{aligned}$$



Proof (Step 2)

Using the following limits for large n :

- $\lim_{n \rightarrow \infty} \frac{n(n-1)(n-2)\dots(n-k+1)}{n^k} = 1$
- $\lim_{n \rightarrow \infty} \left(1 - \frac{\lambda}{n}\right)^n = e^{-\lambda}$
- $\lim_{n \rightarrow \infty} \left(1 - \frac{\lambda}{n}\right)^{-k} = 1$

Thus, we conclude:

$$\lim_{n \rightarrow \infty} P_X(k) = \frac{e^{-\lambda} \lambda^k}{k!}$$



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Cumulative Distribution Function (CDF)

- The CDF is a way to describe the distribution of any random variable (discrete, continuous, or mixed).
- It is defined as:

$$F_X(x) = P(X \leq x), \text{ for all } x \in \mathbb{R}.$$

- The CDF is defined for all $x \in \mathbb{R}$, and is useful for describing both discrete and continuous random variables.



Example: CDF for a Binomial Distribution

Problem: I toss a coin twice. Let X be the number of observed heads. Find the CDF of X .



Solution: CDF for the Binomial Distribution

Problem: I toss a coin twice. Let X be the number of observed heads. Find the CDF of X .

- $X \sim \text{Binomial}(2, \frac{1}{2})$
- The range of X is $R_X = \{0, 1, 2\}$
- The PMF is given by:

$$P_X(0) = \frac{1}{4},$$

$$P_X(1) = \frac{1}{2},$$

$$P_X(2) = \frac{1}{4}.$$



Solution: CDF for the Binomial Distribution

To find the CDF:

- For $x < 0$, $F_X(x) = P(X \leq x) = 0$
- For $x \geq 2$, $F_X(x) = P(X \leq x) = 1$
- For $0 \leq x < 1$, $F_X(x) = P(X \leq x) = P(X = 0) = \frac{1}{4}$
- For $1 \leq x < 2$,
 $F_X(x) = P(X \leq x) = P(X = 0) + P(X = 1) = \frac{1}{4} + \frac{1}{2} = \frac{3}{4}$

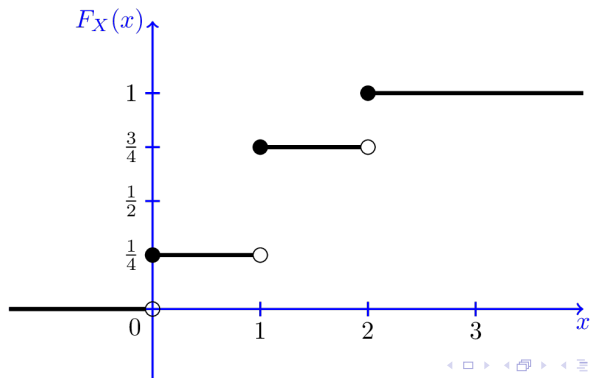
Summary:

$$F_X(x) = \begin{cases} 0 & \text{for } x < 0 \\ \frac{1}{4} & \text{for } 0 \leq x < 1 \\ \frac{3}{4} & \text{for } 1 \leq x < 2 \\ 1 & \text{for } x \geq 2 \end{cases}$$



Graph of CDF

- The CDF is flat between the points in R_X and jumps at each value in the range.
- The size of the jump at each point is equal to the probability at that point.
- For example, at $x = 1$, the CDF jumps from $\frac{1}{4}$ to $\frac{3}{4}$. The size of the jump is $\frac{1}{2}$, which is equal to $P_X(1)$.

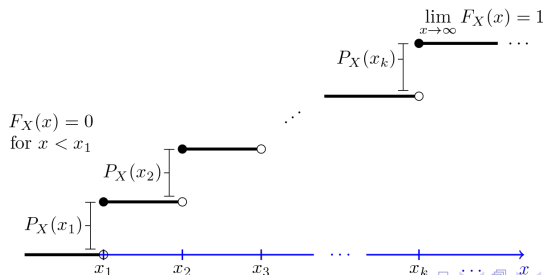


CDF for a General Discrete Random Variable

- Let X be a discrete random variable with range $R_X = \{x_1, x_2, x_3, \dots\}$ such that $x_1 < x_2 < x_3 < \dots$
- Assume x_1 is the smallest value in R_X (if not, $F_X(x) \rightarrow 0$ as $x \rightarrow -\infty$).
- The CDF is given by:

$$F_X(x) = P(X \leq x), \text{ for all } x \in \mathbb{R}.$$

- The CDF for a discrete random variable resembles a staircase, starting at $F_X(-\infty) = 0$ and jumping at each point in the range of X .



Behavior of the CDF

- Between two points in the range, x_k and x_{k+1} , the CDF is constant:

$$F_X(x) = F_X(x_k) \text{ for } x_k \leq x < x_{k+1}.$$

- The CDF jumps at each x_k :

$$F_X(x_k) - F_X(x_k - \epsilon) = P_X(x_k), \quad \text{for small } \epsilon > 0.$$

- The CDF is a non-decreasing function, i.e., if $y \geq x$, then $F_X(y) \geq F_X(x)$.
- As $x \rightarrow \infty$, $F_X(x) \rightarrow 1$.



CDF and PMF Relationship

- The CDF describes the full distribution of a discrete random variable.
- We can obtain the PMF from the CDF by analyzing the jumps:

$$P_X(x_k) = F_X(x_k) - F_X(x_k - \epsilon).$$

- Given the PMF, we can compute the CDF as:

$$F_X(x) = \sum_{x_k \leq x} P_X(x_k).$$



Useful Formula: $P(a < X \leq b)$

For any $a \leq b$: $P(a < X \leq b) = F_X(b) - F_X(a)$

Proof:

- We know that: $P(X \leq b) = P(X \leq a) + P(a < X \leq b)$.
- Therefore: $F_X(b) = F_X(a) + P(a < X \leq b)$.
- Hence, the formula follows.

Important Note

- To find $P(X < x)$ for a discrete random variable, we use:

$$P(X < x) = P(X \leq x) - P(X = x) = F_X(x) - P_X(x).$$

- Be mindful of the difference between $<$ and $\leq$, especially for discrete random variables.



Example: Discrete Random Variable

Let X be a discrete random variable with range $R_X = \{1, 2, 3, \dots\}$. The PMF of X is given by:

$$P_X(k) = \frac{1}{2^k} \text{ for } k = 1, 2, 3, \dots$$

We are tasked to:

- Find and plot the CDF of X , $F_X(x)$.
- Find $P(2 < X \leq 5)$.
- Find $P(X > 4)$.



Finding the CDF $F_X(x)$

First, note that this is a valid PMF because:

$$\sum_{k=1}^{\infty} P_X(k) = \sum_{k=1}^{\infty} \frac{1}{2^k} = 1 \quad (\text{geometric sum}).$$

Next, we find the CDF:

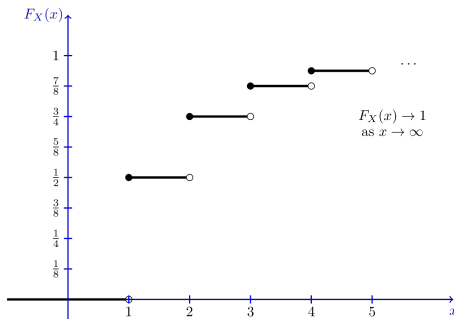
- For $x < 1$, $F_X(x) = 0$.
- For $1 \leq x < 2$, $F_X(x) = P_X(1) = \frac{1}{2}$.
- For $2 \leq x < 3$, $F_X(x) = P_X(1) + P_X(2) = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$.

In general, for $k \leq x < k+1$:

$$F_X(x) = P_X(1) + P_X(2) + \cdots + P_X(k) = \frac{2^k - 1}{2^k}.$$



Finding Probabilities



To find $P(2 < X \leq 5)$: $P(2 < X \leq 5) = F_X(5) - F_X(2) = \frac{31}{32} - \frac{3}{4} = \frac{7}{32}$.

To find $P(X > 4)$:

$$P(X > 4) = 1 - P(X \leq 4) = 1 - F_X(4) = 1 - \frac{15}{16} = \frac{1}{16}.$$



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Expectation (Mean, Average)

- If you have a collection of numbers $a_1, a_2, \dots, a_N$, their average is a single number that describes the whole collection.
- The expected value for a random variable X is defined as the weighted average of the values in the range.

- **Definition of Expected Value**

Let X be a discrete random variable with range $R_X = \{x_1, x_2, x_3, \dots\}$ (finite or countably infinite). The **expected value** of X , denoted by $E[X]$, is defined as:

$$E[X] = \sum_{x_k \in R_X} x_k P(X = x_k) = \sum_{x_k \in R_X} x_k P_X(x_k).$$



Understanding the Concept Behind $E[X]$

Consider a discrete random variable with range $R_X = \{x_1, x_2, x_3, \dots\}$. This random variable results from a random experiment. Suppose we repeat this experiment a very large number of times N , and that the trials are independent.

Let:

- N_1 be the number of times we observe x_1 ,
- N_2 be the number of times we observe x_2 ,
- ...
- N_k be the number of times we observe x_k .

Since $P(X = x_k) = P_X(x_k)$, we expect:

$$P_X(x_1) \approx \frac{N_1}{N}, \quad P_X(x_2) \approx \frac{N_2}{N}, \quad \dots, \quad P_X(x_k) \approx \frac{N_k}{N}.$$

In other words, we have $N_k \approx NP_X(x_k)$.



The Average of Observed Values

Now, if we take the average of the observed values of X , we obtain:

$$\begin{aligned}\text{Average} &= \frac{N_1x_1 + N_2x_2 + N_3x_3 + \dots}{N} \\ &\approx \frac{x_1NP_X(x_1) + x_2NP_X(x_2) + x_3NP_X(x_3) + \dots}{N} \\ &= x_1P_X(x_1) + x_2P_X(x_2) + x_3P_X(x_3) + \dots = E[X].\end{aligned}$$

Thus, the intuition behind $E[X]$ is that if you repeat the random experiment independently N times and take the average of the observed data, the average gets closer and closer to $E[X]$ as N becomes larger and larger.



Notations for Expected Value

We sometimes denote $E[X]$ by μ_X , so different notations for the expected value of X include:

$$E[X] = \mathbb{E}[X] = \mathbb{E}(X) = \mu_X.$$

Let's compute the expected values of some well-known distributions.



Expected Value of Geometric Distribution

Let $X \sim \text{Geometric}(p)$. Find $E[X]$.



Expected Value of Geometric Distribution

Let $X \sim \text{Geometric}(p)$. Find $E[X]$.

Solution: For the geometric distribution, the range is $R_X = \{1, 2, 3, \dots\}$ and the PMF is given by

$$P_X(k) = q^{k-1}p, \quad \text{for } k = 1, 2, \dots$$

where $0 < p < 1$ and $q = 1 - p$. Thus, we can write

$$E[X] = \sum_{x_k \in R_X} x_k P(X = x_k) = \sum_{k=1}^{\infty} kq^{k-1}p = p \sum_{k=1}^{\infty} kq^{k-1}.$$

Using the geometric sum formula:

$$\sum_{k=0}^{\infty} x^k = \frac{1}{1-x}, \quad \text{for } |x| < 1,$$



Expected Value of Geometric Distribution (Cont'd)

we differentiate with respect to x :

$$\frac{d}{dx} \sum_{k=0}^{\infty} x^k = \frac{d}{dx} \frac{1}{1-x} = \frac{1}{(1-x)^2}, \quad \text{for } |x| < 1.$$

Thus, we have

$$\sum_{k=1}^{\infty} kq^{k-1} = \frac{1}{(1-q)^2}.$$

Finally, we can write:

$$E[X] = p \cdot \frac{1}{p^2} = \frac{1}{p}.$$

So, for $X \sim \text{Geometric}(p)$, $E[X] = \frac{1}{p}$.



Intuition Behind the Geometric Expectation

The random experiment behind the geometric distribution is tossing a coin until observing the first heads, where $P(H) = p$. We find that, on average, you need to toss the coin $\frac{1}{p}$ times.

Example: If p is small (heads are unlikely), then $\frac{1}{p}$ is large, so you need to toss the coin many times. Conversely, for large p , fewer tosses usually suffice.



Expected Value of Poisson Distribution (Cont'd)

Let $X \sim \text{Poisson}(\lambda)$. Find $E[X]$.



Expected Value of Poisson Distribution (Cont'd)

Let $X \sim \text{Poisson}(\lambda)$. Find $E[X]$.

Solution: For the Poisson distribution, the range is $R_X = \{0, 1, 2, \dots\}$ and the PMF is given by

$$P_X(k) = \frac{e^{-\lambda} \lambda^k}{k!}, \quad \text{for } k = 0, 1, 2, \dots$$

Thus, we can write:

$$E[X] = \sum_{x_k \in R_X} x_k P(X = x_k) = \sum_{k=0}^{\infty} k \frac{e^{-\lambda} \lambda^k}{k!}.$$

This simplifies to:

$$E[X] = e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^k}{(k-1)!} = e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^{j+1}}{j!} = \lambda e^{-\lambda} e^{\lambda} = \lambda.$$



So, the expected value of a Poisson random variable is $E[X] = \lambda$.

Linearity of Expectation

Theorem: Expectation is linear. For any constants $a, b \in \mathbb{R}$ and random variables $X_1, X_2, \dots, X_n$:

$$E[aX + b] = aE[X] + b,$$

$$E[X_1 + X_2 + \dots + X_n] = E[X_1] + E[X_2] + \dots + E[X_n].$$

This property makes it easier to calculate the expected value of linear functions of random variables.



Expected Value of Binomial Distribution

Let $X \sim \text{Binomial}(n, p)$. Find $E[X]$.

Solution: We provide two ways to solve this problem:

1. Direct Method: We can compute $E[X] = \sum_{x_k \in R_X} x_k P(X = x_k)$, but this requires more algebra.

2. Using Linearity of Expectation: Remember that if $X_1, X_2, \dots, X_n$ are independent Bernoulli(p) random variables, then $X = X_1 + X_2 + \dots + X_n$ has a Binomial(n, p) distribution. Thus:

$$E[X] = E[X_1 + X_2 + \dots + X_n] = E[X_1] + E[X_2] + \dots + E[X_n] =$$

$$p + p + \dots + p = np.$$

So, $E[X] = np$.



Expected Value of Pascal Distribution

Let $X \sim \text{Pascal}(m, p)$. Find $E[X]$.



Expected Value of Pascal Distribution

Let $X \sim \text{Pascal}(m, p)$. Find $E[X]$.

Solution: We claim that if $X_1, X_2, \dots, X_m$ are independent and $X_i \sim \text{Geometric}(p)$, then $X = X_1 + X_2 + \dots + X_m$ has a $\text{Pascal}(m, p)$ distribution. Since we know $E[X_i] = \frac{1}{p}$, we conclude:

$$\begin{aligned} E[X] &= E[X_1 + X_2 + \dots + X_m] \\ &= E[X_1] + E[X_2] + \dots + E[X_m] \\ &= \frac{1}{p} + \frac{1}{p} + \dots + \frac{1}{p} = \frac{m}{p}. \end{aligned}$$

Thus, $E[X] = \frac{m}{p}$.



Functions of Random Variables

If X is a random variable and $Y = g(X)$, then Y itself is a random variable. Thus, we can talk about its PMF, CDF, and expected value. The range of Y can be written as:

$$R_Y = \{g(x) | x \in R_X\}.$$

If we already know the PMF of X , to find the PMF of $Y = g(X)$, we can write:

$$P_Y(y) = P(Y = y) = P(g(X) = y) = \sum_{x: g(x)=y} P_X(x).$$



Example: Finding PMF of a Transformed Variable

Let X be a discrete random variable with $P_X(k) = \frac{1}{5}$ for $k = -1, 0, 1, 2, 3$.
Let $Y = 2|X|$. Find the range and PMF of Y .



Example: Finding PMF of a Transformed Variable

$P_X(k) = \frac{1}{5}$ for $k = -1, 0, 1, 2, 3$. Let $Y = 2|X|$. Range and PMF of Y .

Solution:

First, the range of Y is:

$$R_Y = \{2|x| \text{ where } x \in R_X\} = \{0, 2, 4, 6\}.$$

Now, to find $P_Y(y)$, we compute $P(Y = y)$ for $y = 0, 2, 4, 6$.

$$P_Y(0) = P(Y = 0) = P(2|X| = 0) = P(X = 0) = \frac{1}{5},$$

$$P_Y(2) = P(Y = 2) = P(2|X| = 2) = P(X = -1 \text{ or } X = 1) = P_X(-1) + P_X(1) = \frac{1}{5} + \frac{1}{5} = \frac{2}{5},$$

$$P_Y(4) = P(Y = 4) = P(2|X| = 4) = P(X = 2) = \frac{1}{5},$$

$$P_Y(6) = P(Y = 6) = P(2|X| = 6) = P(X = 3) = \frac{1}{5}.$$

So, the PMF of Y is summarized as:

$$P_Y(k) = \begin{cases} \frac{1}{5} & \text{for } k = 0, 4, 6, \\ \frac{2}{5} & \text{for } k = 2, \\ 0 & \text{otherwise.} \end{cases}$$



Expected Value of a Function of a Random Variable (LOTUS)

Let X be a discrete random variable with PMF $P_X(x)$, and let $Y = g(X)$. To find $E[Y]$, one way is to find the PMF of Y and then use the expectation formula:

$$E[Y] = E[g(X)] = \sum_{y \in R_Y} y P_Y(y).$$

However, a simpler method is the Law of the Unconscious Statistician (LOTUS):

$$E[g(X)] = \sum_{x_k \in R_X} g(x_k) P_X(x_k).$$



Example: Finding $E[\sin(X)]$

Let X be a discrete random variable with range $R_X = \{0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi\}$ and $P_X(0) = P_X(\frac{\pi}{4}) = P_X(\frac{\pi}{2}) = P_X(\frac{3\pi}{4}) = P_X(\pi) = \frac{1}{5}$. Find $E[\sin(X)]$.



Example: Finding $E[\sin(X)]$

Let X be a discrete random variable with range $R_X = \{0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi\}$ and $P_X(0) = P_X(\frac{\pi}{4}) = P_X(\frac{\pi}{2}) = P_X(\frac{3\pi}{4}) = P_X(\pi) = \frac{1}{5}$. Find $E[\sin(X)]$.

Solution: Using LOTUS, we have:

$$E[\sin(X)] = \sum_{x_k \in R_X} \sin(x_k) P_X(x_k).$$

Substituting values:

$$E[\sin(X)] = \sin(0) \cdot \frac{1}{5} + \sin\left(\frac{\pi}{4}\right) \cdot \frac{1}{5} + \sin\left(\frac{\pi}{2}\right) \cdot \frac{1}{5} + \sin\left(\frac{3\pi}{4}\right) \cdot \frac{1}{5} + \sin(\pi) \cdot \frac{1}{5}.$$

Using trigonometric values:

$$E[\sin(X)] = 0 \cdot \frac{1}{5} + \frac{\sqrt{2}}{2} \cdot \frac{1}{5} + 1 \cdot \frac{1}{5} + \frac{\sqrt{2}}{2} \cdot \frac{1}{5} + 0 \cdot \frac{1}{5}$$

$$E[\sin(X)] = \frac{\sqrt{2}}{10} + \frac{1}{5} + \frac{\sqrt{2}}{10} = \frac{2\sqrt{2}}{10} + \frac{1}{5} = \frac{\sqrt{2}}{5} + \frac{1}{5}.$$



Linearity of Expectation (Proof)

Example: Prove that $E[aX + b] = aE[X] + b$.

Solution: Here, $g(X) = aX + b$, so using LOTUS we have:

$$E[aX + b] = \sum_{x_k \in R_X} (ax_k + b)P_X(x_k).$$

This can be split into:

$$E[aX + b] = \sum_{x_k \in R_X} ax_k P_X(x_k) + \sum_{x_k \in R_X} bP_X(x_k),$$

$$E[aX + b] = a \sum_{x_k \in R_X} x_k P_X(x_k) + b \sum_{x_k \in R_X} P_X(x_k),$$

$$E[aX + b] = aE[X] + b.$$

Thus, we have proved that $E[aX + b] = aE[X] + b$.



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Variance

Consider two random variables X and Y with the following PMFs:

$$P_X(x) = \begin{cases} 0.5 & \text{for } x = -100, \\ 0.5 & \text{for } x = 100, \\ 0 & \text{otherwise.} \end{cases} \quad P_Y(y) = \begin{cases} 1 & \text{for } y = 0, \\ 0 & \text{otherwise.} \end{cases}$$

Note that $E[X] = E[Y] = 0$. Although both random variables have the same mean value, their distributions are completely different. Y is always equal to its mean of 0, while X is either 100 or -100, quite far from its mean.

The variance is a measure of how spread out the distribution of a random variable is. Here, the variance of Y is quite small since its distribution is concentrated at a single value, while the variance of X will be larger since its distribution is more spread out.



Definition of Variance

- The variance of a random variable X , with mean $E[X] = \mu_X$, is defined as:

$$\text{Var}(X) = E[(X - \mu_X)^2].$$

- By definition, the variance of X is the average value of $(X - \mu_X)^2$. Since $(X - \mu_X)^2 \geq 0$, the variance is always larger than or equal to zero.
- A large value of the variance means that $(X - \mu_X)^2$ is often large, so X often takes values far from its mean. This indicates a spread-out distribution. Conversely, a low variance means the distribution is concentrated around its mean.
- Note that if we did not square the difference between X and its mean, the result would be 0. That is,

$$E[X - \mu_X] = E[X] - E[\mu_X] = \mu_X - \mu_X = 0.$$



Computing Variance using LOTUS

To compute $\text{Var}(X) = E[(X - \mu_X)^2]$, note that we need to find the expected value of $g(X) = (X - \mu_X)^2$. Using LOTUS, we can write:

$$\text{Var}(X) = E[(X - \mu_X)^2] = \sum_{x_k \in R_X} (x_k - \mu_X)^2 P_X(x_k).$$

For example, for X and Y defined earlier, we have:

$$\text{Var}(X) = (-100 - 0)^2(0.5) + (100 - 0)^2(0.5) = 10,000,$$

$$\text{Var}(Y) = (0 - 0)^2(1) = 0.$$

As expected, X has a large variance, while $\text{Var}(Y) = 0$.



Standard Deviation

Note that $\text{Var}(X)$ has a different unit than X . For example, if X is measured in meters, then $\text{Var}(X)$ is in square meters (meters^2). To address this, we define the standard deviation, denoted by σ_X , which is the square root of the variance.

The standard deviation of a random variable X is defined as:

$$\text{SD}(X) = \sigma_X = \sqrt{\text{Var}(X)}.$$

The standard deviation of X has the same unit as X . For X and Y defined earlier, we have:

$$\sigma_X = \sqrt{10,000} = 100,$$

$$\sigma_Y = \sqrt{0} = 0.$$



Computational Formula for Variance

A useful formula for computing the variance is:

$$\text{Var}(X) = E[X^2] - (E[X])^2 \quad (3.5)$$

To prove this, note that:

$$\text{Var}(X) = E[(X - \mu_X)^2] = E[X^2 - 2\mu_X X + \mu_X^2] = E[X^2] - 2E[\mu_X X] + E[\mu_X^2]$$

By linearity of expectation, and since μ_X is a constant, we get:

$$E[\mu_X X] = \mu_X E[X] = \mu_X^2$$

Thus,

$$\text{Var}(X) = E[X^2] - \mu_X^2$$



Computational Formula for Variance

Equation (3.5) is often easier to work with compared to $\text{Var}(X) = E[(X - \mu_X)^2]$. To use this equation, we can find $E[X^2]$ using LOTUS:

$$E[X^2] = \sum_{x_k \in R_X} x_k^2 P_X(x_k),$$

and then subtract μ_X^2 to obtain the variance.



Example: Fair Die

Example: I roll a fair die and let X be the resulting number. Find $E[X]$, $\text{Var}(X)$, and σ_X .



Example: Fair Die

Example: I roll a fair die and let X be the resulting number. Find $E[X]$, $\text{Var}(X)$, and σ_X .

Solution: We have $R_X = \{1, 2, 3, 4, 5, 6\}$ and $P_X(k) = \frac{1}{6}$ for $k = 1, 2, \dots, 6$. Thus, we have:

$$E[X] = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6} = \frac{7}{2}$$

$$E[X^2] = 1^2 \cdot \frac{1}{6} + 2^2 \cdot \frac{1}{6} + 3^2 \cdot \frac{1}{6} + 4^2 \cdot \frac{1}{6} + 5^2 \cdot \frac{1}{6} + 6^2 \cdot \frac{1}{6} = \frac{91}{6}$$

Thus,

$$\text{Var}(X) = E[X^2] - (E[X])^2 = \frac{91}{6} - \left(\frac{7}{2}\right)^2 = \frac{91}{6} - \frac{49}{4} \approx 2.92,$$

and

$$\sigma_X = \sqrt{\text{Var}(X)} \approx \sqrt{2.92} \approx 1.71.$$



Linearity of Variance

Note that variance is not a linear operator. In particular, we have the following theorem:

Theorem: For a random variable X and real numbers a and b ,

$$\text{Var}(aX + b) = a^2 \text{Var}(X) \quad (3.6)$$

Proof: If $Y = aX + b$, then $E[Y] = aE[X] + b$. Thus,

$$\begin{aligned} \text{Var}(Y) &= E[(Y - E[Y])^2] \\ &= E[(aX + b - aE[X] - b)^2] \\ &= E[a^2(X - \mu_X)^2] \\ &= a^2 E[(X - \mu_X)^2] \\ &= a^2 \text{Var}(X) \end{aligned}$$



Standard Deviation and Sum of Independent Variables

From Equation (3.6), we conclude that, for standard deviation,

$$\text{SD}(aX + b) = |a|\text{SD}(X).$$

While variance is not a linear operation, there is a very important case where it behaves like a linear operator: when we look at the sum of independent random variables.

Theorem: If $X_1, X_2, \dots, X_n$ are independent random variables and $X = X_1 + X_2 + \dots + X_n$, then:

$$\text{Var}(X) = \text{Var}(X_1) + \text{Var}(X_2) + \dots + \text{Var}(X_n) \quad (3.7)$$



Example: Binomial Distribution

Example: If $X \sim \text{Binomial}(n, p)$, find $\text{Var}(X)$.

Solution: We know that a Binomial random variable can be written as the sum of n independent Bernoulli random variables, i.e.,
 $X = X_1 + X_2 + \cdots + X_n$. Thus, we conclude:

$$\text{Var}(X) = \text{Var}(X_1) + \text{Var}(X_2) + \cdots + \text{Var}(X_n).$$

If $X_i \sim \text{Bernoulli}(p)$, then its variance is:

$$\text{Var}(X_i) = E[X_i^2] - (E[X_i])^2 = 1^2 \cdot p + 0^2 \cdot (1 - p) - p^2 = p(1 - p).$$

Thus,

$$\text{Var}(X) = p(1 - p) + p(1 - p) + \cdots + p(1 - p) = np(1 - p)$$



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Problem 1

Let X be a discrete random variable with the following PMF:

$$P_X(x) = \begin{cases} 0.3 & \text{for } x = 3 \\ 0.2 & \text{for } x = 5 \\ 0.3 & \text{for } x = 8 \\ 0.2 & \text{for } x = 10 \\ 0 & \text{otherwise} \end{cases}$$

Find and plot the CDF of X .



Solution Problem 1

The CDF is defined by $F_X(x) = P(X \leq x)$. We have

$$F_X(x) = \begin{cases} 0 & \text{for } x < 3 \\ P_X(3) = 0.3 & \text{for } 3 \leq x < 5 \\ P_X(3) + P_X(5) = 0.5 & \text{for } 5 \leq x < 8 \\ P_X(3) + P_X(5) + P_X(8) = 0.8 & \text{for } 8 \leq x < 10 \\ 1 & \text{for } x \geq 10 \end{cases}$$



Problem 2: Finding $E[X]$, $\text{Var}(X)$, and $E[Y]$

Let X be a discrete random variable with the following PMF:

$$P_X(k) = \begin{cases} 0.1 & \text{for } k = 0 \\ 0.4 & \text{for } k = 1 \\ 0.3 & \text{for } k = 2 \\ 0.2 & \text{for } k = 3 \\ 0 & \text{otherwise} \end{cases}$$

Find $E[X]$, $\text{Var}(X)$, and $E[(X - 2)^2]$.



Solution Problem 2

$$E[X] = \sum_{k \in R_X} kP_X(k) = 0(0.1) + 1(0.4) + 2(0.3) + 3(0.2) = 1.6$$

Next, find $\text{Var}(X)$ using $\text{Var}(X) = E[X^2] - (E[X])^2$.

We need $E[X^2]$:

$$E[X^2] = 0^2(0.1) + 1^2(0.4) + 2^2(0.3) + 3^2(0.2) = 3.4$$

$$\text{Var}(X) = 3.4 - (1.6)^2 = 0.84$$

$$E[(X-2)^2] = (0-2)^2(0.1) + (1-2)^2(0.4) + (2-2)^2(0.3) + (3-2)^2(0.2) = 1$$



Problem 3: Finding the PMF of $Y = X(X - 1)(X - 2)$

Let X be a discrete random variable with PMF:

$$P_X(k) = \begin{cases} 0.2 & \text{for } k = 0 \\ 0.2 & \text{for } k = 1 \\ 0.3 & \text{for } k = 2 \\ 0.3 & \text{for } k = 3 \\ 0 & \text{otherwise} \end{cases}$$

Define $Y = X(X - 1)(X - 2)$. Find the PMF of Y .



Solution Problem 3

First, note that $R_Y = \{x(x-1)(x-2) \mid x \in \{0, 1, 2, 3\}\} = \{0, 6\}$.

$$P_Y(0) = P(X = 0 \text{ or } X = 1 \text{ or } X = 2) = P_X(0) + P_X(1) + P_X(2) = 0.7$$

$$P_Y(6) = P(X = 3) = 0.3$$

Thus, the PMF of Y is:

$$P_Y(k) = \begin{cases} 0.7 & \text{for } k = 0 \\ 0.3 & \text{for } k = 6 \\ 0 & \text{otherwise} \end{cases}$$



Problem 4

Let $X \sim \text{Geometric}(p)$. Find $\mathbb{E} \left[\frac{1}{2^X} \right]$.



Solution Problem 4

The PMF of X is given by:

$$P_X(k) = \begin{cases} pq^{k-1} & \text{for } k = 1, 2, 3, \dots \\ 0 & \text{otherwise} \end{cases}$$

where $q = 1 - p$. Thus,

$$\begin{aligned} \mathbb{E} \left[\frac{1}{2^X} \right] &= \sum_{k=1}^{\infty} \frac{1}{2^k} P_X(k) \\ &= \sum_{k=1}^{\infty} \frac{1}{2^k} pq^{k-1} \\ &= \frac{p}{2} \sum_{k=1}^{\infty} \left(\frac{q}{2} \right)^{k-1} \\ &= \frac{p}{2} \frac{1}{1 - \frac{q}{2}} = \frac{p}{2} \cdot \frac{2}{2 - q} = \frac{p}{1 + p} \end{aligned}$$



Problem 5

If $X \sim \text{Hypergeometric}(b, r, k)$, find $\mathbb{E}[X]$.



Solution Problem 5

If $X \sim \text{Hypergeometric}(b, r, k)$, find $\mathbb{E}[X]$.

Solution:

The PMF of X is given by:

$$P_X(x) = \begin{cases} \frac{\binom{b}{x} \binom{r}{k-x}}{\binom{b+r}{k}}, & x \in R_X \\ 0, & \text{otherwise} \end{cases}$$

where $R_X = \{\max(0, k - r), \max(0, k - r) + 1, \dots, \min(k, b)\}$.

Instead of finding $\mathbb{E}[X]$ directly, we use the linearity of expectation. Let X_i be the indicator random variable for the i th chosen marble being blue:

$$X = X_1 + X_2 + \dots + X_k.$$

Thus,

$$\mathbb{E}[X] = \mathbb{E}[X_1] + \mathbb{E}[X_2] + \dots + \mathbb{E}[X_k].$$



Solution Problem 5

Since all marbles are equally likely to be chosen, we have:

$$P(X_i = 1) = \frac{b}{b+r}.$$

Therefore,

$$\mathbb{E}[X_i] = \frac{b}{b+r}.$$

Thus,

$$\mathbb{E}[X] = k \frac{b}{b+r}.$$



Thanks

